



```

rain_count = 0

def wind_callback(channel):
    global wind_count
    wind_count += 1

def rain_callback(channel):
    global rain_count
    rain_count += 1

GPIO.add_event_detect(ANEMOMETER_PIN, GPIO.FALLING,
    callback=wind_callback,
    bouncetime=300)
GPIO.add_event_detect(RAIN_GAUGE_PIN, GPIO.FALLING,
    callback=rain_callback,
    bouncetime=300)

def read_dht22():
    humidity, temperature = Adafruit_DHT.read_retry(DHT_SENSOR, DHT_PIN)
    return temperature, humidity

def read_bmp280():
    pressure = bmp280.read_pressure() / 100.0
    return pressure

def read_sds011():
    pm25, pm10 = sds011.query()
    return pm25, pm10

def read_anemometer():
    global wind_count
    wind_speed = wind_count * 1.2 / 5.0 # m/s
    wind_count = 0
    return wind_speed

def read_rain_gauge():
    global rain_count
    rainfall = rain_count * 0.2794 # mm
    rain_count = 0
    return rainfall

if name == "main":
    try:
        while True:
            temperature, humidity = read_dht22()
            pressure = read_bmp280()
            pm2_5, pm10 = read_sds011()
            wind_speed = read_anemometer()
            rainfall = read_rain_gauge()

            # Process the data and display or store it
            print(f'Temperature: {temperature} C, Humidity: {humidity}%, Pressure: {pressure} hPa')
            print(f'PM2.5: {pm2_5} ug/m3, PM10: {pm10} ug/m3')
            print(f'Wind speed: {wind_speed} m/s')
            print(f'Rainfall: {rainfall} mm')

            time.sleep(60)
    except KeyboardInterrupt:
        GPIO.cleanup()

```

Section 6: Operation Instructions

1. Power on the Raspberry Pi and ensure it is connected to the internet.
2. Transfer the Python script (weather_station.py) to the Raspberry Pi using an SCP client or other method.
3. Make the script executable using the following command:
`chmod +x weather_station.py`
4. Run the script:
`./weather_station.py`
5. The script will start reading data from the sensors and display it on the console. You can modify the code to store the data in a file or database, or visualize it using a web interface.
6. To make the script run automatically at startup, edit the /etc/rc.local file and add the following line before the exit 0 line:
`/path/to/weather_station.py &`

Next steps...

Congratulations! You have successfully built a fully functional weather station using a Raspberry Pi 4 Model B board. With this weather station, you can monitor temperature, humidity, atmospheric pressure, air quality, wind speed, wind gusts, wind direction, and rainfall in real-time. You can further customise the project by adding more sensors, integrating it with smart home systems, or creating a web interface for remote monitoring. The possibilities are endless!

Keep in mind that the provided code is a basic implementation, and you may need to adjust the calculations and values based on your specific sensors and their datasheets. For instance, you might need to calibrate the anemometer and rain gauge to provide accurate readings.

Additionally, you can expand this project to visualise the collected data in a more user-friendly way. Consider creating a web application using Flask or Django to display real-time sensor data or historical trends.

To store the data, you can use SQLite, a lightweight database that doesn't require a separate server process. Alternatively, you can use more advanced databases like MySQL or PostgreSQL if you plan to scale up your project or if you want to implement a more sophisticated data analysis system.

To ensure your weather station provides accurate data over time, regular maintenance, such as cleaning the sensors and ensuring they are protected from the elements, is essential. Depending on your specific location and climate, you may need to adjust the weatherproof enclosure to protect the components from extreme temperatures, humidity, or other environmental factors.

By following this guide and applying some creativity, you can build a weather station tailored to your specific needs and enjoy the benefits of having real-time, personalised weather data at your fingertips. 📌

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